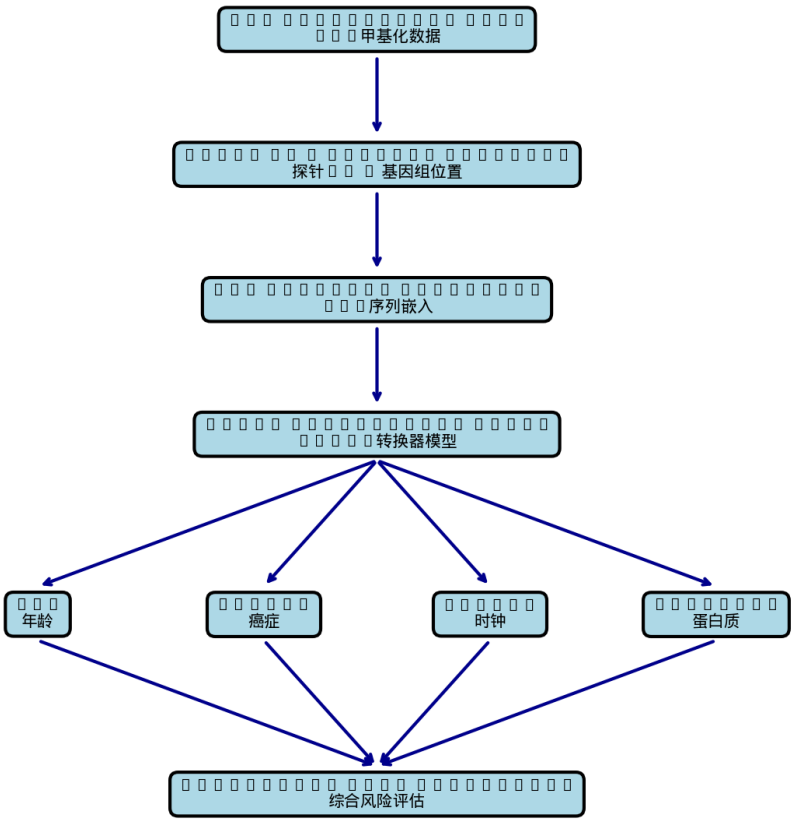
DNA Methylation Analysis Report

DNA■■■■■■■

1. Executive Summary / ■■■■

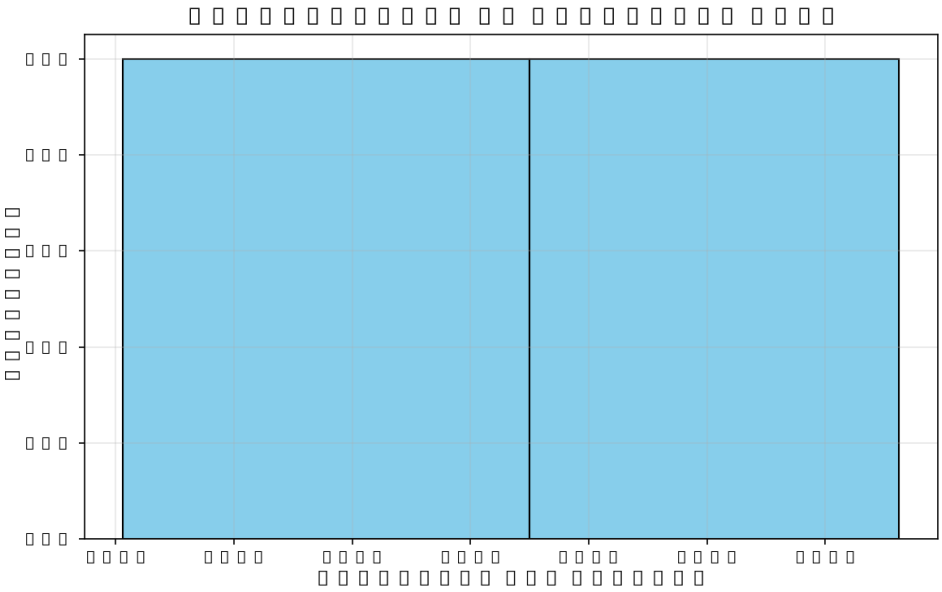
Metric / ■■	Value / ■
Total Samples / ■■■■	2
Average Predicted Age / ■■■■■■	59.2 years
Cancer Detection Rate / ■■■■■	100.0%
Average GrimAge2 / ■■GrimAge2	65.4 years

2. Prediction Methodology / ■■■■■

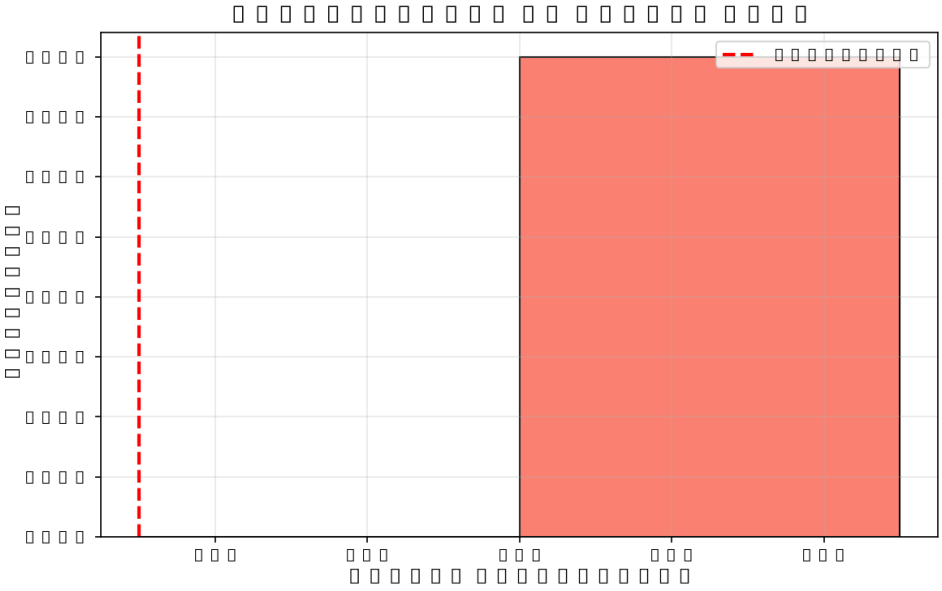


3. Detailed Results / ■■■■

3.1 Age Distribution / ■■■■



3.2 Cancer Risk Distribution / ■■■■■■

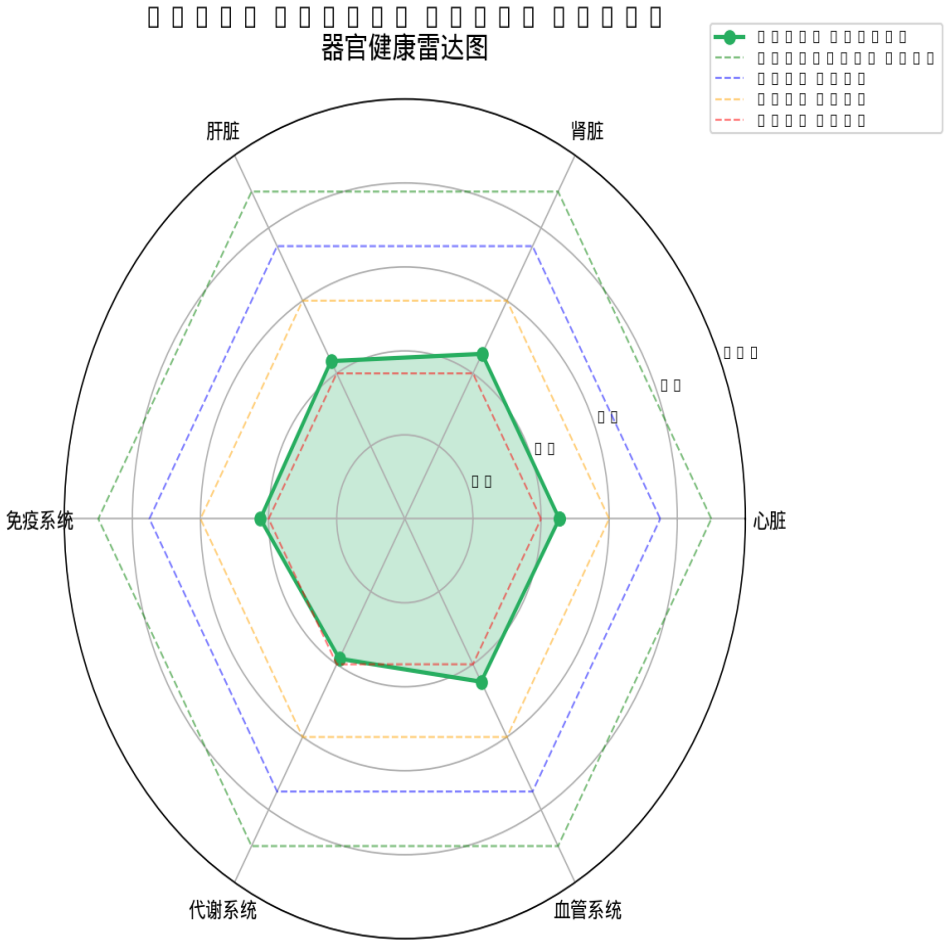


4. Risk Stratification Summary / ■■■■■■

Risk Category / ■■■■	Count / ■■	Percentage / ■■■
■■■	2	100.0%

5. Organ Health Scores / ■■■■■■

Organ System / ■■■■	Average Score / ■■■■alth Level / ■■■■
■■ (Heart)	45.3	■■
■■ (Kidney)	45.4	■■
■■ (Liver)	43.4	■■
■■■■ (Immune System)	42.4	■■
■■■■ (Metabolic System)	38.5	■
■■■■ (Vascular System)	44.8	■■
■■■■ (Overall)	**43.3**	■■



Organ Health Score Interpretation / ■■■■■■■■■
- **90-100 (■■/Excellent):** Optimal organ function / ■■■■■■
- **75-89 (■■/Good):** Good organ health / ■■■■■■
- **60-74 (■■/Fair):** Moderate concerns / ■■■■■■
- **40-59 (■■/Poor):** Significant concerns / ■■■■■■
- **0-39 (■/Very Poor):** Serious concerns / ■■■■■■

Note: Scores are based on protein biomarkers predicted from DNA methylation data. Lower protein levels generally indicate better health (less inflammation, better function).

■■■ ■■■■■DNA■■■■■■■■■■■■■■■■■■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■

6. Methodology Notes / [redacted]

CpGPT Model: A transformer-based deep learning model trained on DNA methylation data to predict biological age, disease risk, and protein levels.

CpGPT[redacted] [redacted]Transformer[redacted]DNA[redacted]
[redacted]

Epigenetic Clocks:
- **GrimAge2:** Mortality predictor based on DNAm surrogates of plasma proteins
- **DunedinPACE:** Pace of aging measure
- **PhenoAge:** Phenotypic age predictor
- **Horvath:** Pan-tissue age predictor

[redacted]
- **GrimAge2:** [redacted]DNAm[redacted]
- **DunedinPACE:** [redacted]
- **PhenoAge:** [redacted]
- **Horvath:** [redacted]

Organ Health Scores: Based on organ-specific protein biomarkers predicted from DNA methylation. Scores reflect the functional status of major organ systems including heart, kidney, liver, immune system, metabolic system, and vascular system.

[redacted] [redacted]DNA[redacted]
[redacted]

Risk Stratification: Based on protein biomarkers, epigenetic clocks, and cancer prediction models.

[redacted] [redacted]

References:
- Nature 2023: Organ aging signatures in the plasma proteome
- Lancet Digital Health 2025: Proteomic organ-specific ageing signatures